

Improving Fraud Detection and Concept Drift Adaptation in Credit Card Transactions Using Incremental Gradient Boosting Trees

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Abstract—Due to the increase in the use of credit cards in electronic shopping, card payments for online commerce have rapidly become a popular trend, which also led to the growth in the number of retailers. Because of these various online shopping options, a more frequent variation in the spending behaviors of the customers and purchasing trends of online markets, known as concept drift problem, can be observed over time which also causes an increase in the need for novel fraudulent strategies. This drifting problem may significantly hinder the effective performance of state-of-the-art fraud detection approaches in real credit card transaction data, which also has the imbalanced class distribution problem. In this study, a card-based incremental Gradient Boosting Tree (GBT) is investigated to detect credit card frauds and to adapt in real-time to drifts occurred in online transactions. The card-based incremental learning is achieved in which the transactions of the fraudulent credit cards reported in each day are incrementally learned by the GBT model. Therefore, the card-based incremental GBT model is compared with the regular GBT model, and retraining of a new transaction set formed by combining the previous set and the transactions of the cards reported as fraudulent. The experiments have been carried out on the 4-month real transaction data from December 2019 to March 2020 in which the concept drift problem occurred in December, dramatically affecting the performance of the GBT model. In these experiments, the improvements in the fraud detection performance have been realized in all months, and also the effectiveness of the card-based increment has been verified by comparing it with the transaction-based incremental learning that may cause *catastrophic forgetting* problem.

Index Terms—Fraud detection, Concept Drift, Imbalanced Data, Gradient Boosting Trees, Incremental Learning, Catastrophic Forgetting.

I. INTRODUCTION

With the rapid growth in information technology and electronic commerce, the use of credit cards for online purchases has raised in recent years. It is expected that the loss will exceed \$35 billion in 2020 according to the Nilson report 2016 [1]. Besides, in 2018, the loss of the world-wise credit card fraud amounted to \$27.85 billion in which 54% of the loss arises from Card-Not-Present (CNP) transactions according to the Nilson report 2019 [2]. To detect the frauds in the streaming online transactions, the financial institutions utilize a rule-based system that requires costly human efforts. To reduce

the loss and to improve the performance of fraud detection, machine learning algorithms are employed on real credit card transactions. However, the streaming real transactions prone to non-stationarity, which involves one of the most-known problems that is sudden changes in concepts such as classes or non-class attributes, known as concept drift [3], [4]. The unexpected drifts in credit card transaction data are often observed due to changes in behaviors or habits of customers, strategies of fraudsters to bypass fraud detection systems, spending behaviors in seasonal or special occasions, etc. Also, the trends of online markets differ over time, such as creating a new payment channel, a new merchant, or a new category for merchants. Traditional fraud detection models are trained using historical transactions, and cannot adapt to concept changes, so the fraud detection performance is expected to dramatically degrade over time.

In addition to the concept drift problem, another common issue of real-world binary classification tasks is imbalanced class distribution. In these problems, minority classes in the data are much more essential, because a wrongly predicted instance of the minority class may cause much higher costs as observed in credit card fraud detection problem. Therefore, it is more important to provide high accuracy on the prediction of the samples of the minority class. However, state-of-the-art classification algorithms tend to be biased toward the majority classes [5], and so provide low accuracy on the minority ones. For this purpose, various approaches have been developed for oversampling on minority classes or undersampling on majority classes. However, undersampling may result in large computational costs and may not guarantee to improve the performance on real data, whereas oversampling may result in loss of important information, as well as not guaranteeing high performance and not solving the problem of imbalance [6]. Therefore, we utilized a method for automatic training set construction presented in the previous work [7] which is a card-based equalization on fraud and non-fraud cards considering a time-period with a number of consecutive months.

Furthermore, the concept drift problem in the imbalanced data, which is a remarkably complicated problem, is still an

unexplored issue. The credit card transactions usually form an imbalanced data set, including extremely few fraud transactions in comparison with the legitimate ones [8]. The gradient boosting tree (GBT), which aims to ensemble weak decision trees in a sequential manner while gradually minimizing a loss function, is a widely used machine learning method for binary classification on imbalanced data is used in the real transactions with real concept drifts. In this work, to adapt to the drifted transactions, the GBT model is evolved by incrementally learning the additional transactions in real-time. At the end of a day, IDs of the credit cards of the transactions reported as a fraud in the last 24-hour are fetched from the related database. Thereafter, all the fraud or legitimate transactions of these fraudulent cards not included in the initial training set, in the same time period used in the construction of the initial set are incrementally learned by the current GBT model. In the real-time learning process, the ensemble is updated based on the new transactions by adding new decision trees to form a stronger GBT model, considering a loss function. Also, retraining all the transactions including the transactions of the reported cards is investigated. However, the memory and computation time required for retraining are so expensive, but a similar improvement in the performances of the fraud detection is observed with the training time that is almost three thousand times faster than the retraining.

Moreover, the main problem of incremental learning known as catastrophic forgetting [9], which is to lose previously learned knowledge is investigated on the real transactions. The forgetting problem is observed while incrementally learning only these daily transactions. To preserve the previous knowledge about the legitimate and fraudulent transactions, the card-based incremental learning method is employed, which requires less than half a second in a day. Using this incremental method, we showed that the detection of credit card fraud and the learning of new transactions can be robustly achieved in real-time. Also, the poor performance of the regular GBT model on the transactions of December with the concept drift problem has been significantly increased by this proposed incremental method. Briefly, using the card-based incremental GBT model, our contributions are (i) learning in real-time for fraud detection on the real bank transactions, (ii) the adaptation to the drifted transactions, (iii) substantial reduction of the retraining time required for this adaptation, (iv) the significant improvement of false positive rate (FPR) and recall of the fraud detection, (v) discarding the use of the automatic training set construction module proposed in the previous work [7], and (vi) overcoming the catastrophic forgetting problem that can be observed in the real-time learning.

II. RELATED WORK

In this section, apart from providing a brief overview of the aforementioned concept drift problem with the imbalanced data problem, we also scrutinize the incremental learning methods to adapt to the drifting problem in credit card fraud detection task.

In several studies, new computational methods have been developed for the detection of concept drift or adaptive to the drift problem in the fraud detection task. However, some of them have mainly focused on the drift problem, especially in real credit card transactions. In [10], a method with a sliding window and an ensemble of classifiers has been proposed to overcome this drifting problem in the real credit card transactions. Another fraud detection based on a concept drift management approach was developed to retain new concepts on the transaction streams using the cardholder profiles [11]. However, the work has been evaluated on synthetic data due to the unavailability of the real credit card data and benchmark data sets.

The boosting algorithms have been utilized for various binary classification tasks such as credit card fraud detection. For instance, [12] used a light gradient boosting machine (LightGBM) model for detecting credit card frauds. In [13], for the detection of credit card frauds, the boosting algorithms, namely, AdaBoost, gradient boost, and eXtreme Gradient Boosting (XGBoost), were implemented to evaluate the detection performances. It was found out that the XGBoost method outperforms the other algorithms. Also, [14] showed how an XGBoost-based solution outperforms three state-of-the-art algorithms that are support vector machine, random forest, and logistic regression.

However, there is only a few work utilizing the incremental learning capability of the GBT algorithm. [15] worked on incremental GBT learning in which only new samples are utilized to update the existing GBT model on the task of plagiarism detection. In [8], the incremental learning problem is discussed under the concept drift problem by comparing various incremental algorithms including the proposed one, on different concept drift data sets with varied distribution ratios between the classes. However, these algorithms have been evaluated on the synthetic data sets.

However, there is a few works in the literature about an incremental model for the detection of credit card frauds. To handle the imbalance problem, an incremental automated data balancing approach was proposed in [16] to improve the fraud detection performance on massive data. In this work, we have also applied card-level undersampling by equalizing fraudulent and legitimate credit card counts while making sure that the transactions of each credit card cover the equal time range. An incremental solution that is based on weighting bags created with transaction windows on transaction data stream was developed to handle the drifts gradually occurring due to class and non-class attribute drifts and imbalance problems [17]. In [18], an unsupervised method based on fuzzy *c*-mean has been developed for incremental learning of the credit card transactions. For the credit card threat assessment, an ensemble incremental framework was introduced in [19], called NIK algorithm composed of three base incremental learners. A batch learner is generated on a new incoming credit data, and then ensembled with the other learners through a voting technique. However, our work based on an incremental learning method is a novel solution for concept drift problem

observed in the real bank transactions.

III. PROPOSED INCREMENTAL FRAUD DETECTION METHOD

Our incremental GBT method for the detection of credit card frauds is composed of two main modules. (i) offline training and (ii) real-time detection as shown in Fig. 1. There is a feature engineering process common in all of the modules. The offline training phase covers the automatic generation of the training set to be used in the generation of a model for fraud detection. In the real-time detection phase, the feature vector is extracted from streaming transactions to the GBT model to decide whether it is fraudulent or not.

A. Feature Engineering

A feature engineering process developed in the previous work [7] to encode the raw categorical features, extracting the embedding features from the textual information like merchant name, and aggregate raw features such as currency, amount, merchant category on the previous transactions of a credit card in a predefined short time range to extract the customer's online purchasing behavior patterns is required before an initial model generation, real-time fraud detection, and learning in real-time. The feature engineering process aims to improve the performance of the model under the imbalanced class distributions, the noisy labels and features, skewness of the data, and overlapping transactions existing in both of the classes. A combination of the different types of features, selection among the distinctive features, and generation of the additional hand-crafted features are used in the feature engineering process. The feature engineering is performed in each of the modules in the fraud detection method, which are the initial model generation, real-time detection, and incremental learning processes.

B. Incremental Learning

In this work, the incremental learning capability is utilized to overcome the concept drift problem to adapt to the unexpected changes in the streaming real transaction data. Firstly, an initial training set is automatically constructed by equalizing fraudulent and legitimate credit card counts and making sure that the transactions of each credit card cover the equal time range. Instead of equalizing fraud and non-fraud transaction count, the same number of training instances for fraud and non-fraud transactions are utilized to train an initial GBT model. After the generation of an initial GBT model, at the end of the day, all the transactions in the same time range belonging to the cards of the known fraudulent transactions are incrementally learned by ensembling additional one decision tree into the recent model. However, in each day, only a couple of hundred fraudulent transactions are reported, and incrementally inserting the decision tree based only on these frauds may cause catastrophic forgetting, or may not learn anything about the fraudulent or non-fraudulent features. Therefore, all the transactions of the fraudulent cards, including the legitimate and fraudulent transactions are utilized in the real-time incremental learning process.

TABLE I: Overview of the data sets for December 2019.

Overview	Training Set Feb.'19-Nov.'19	Test Set Dec.'19
Number of Transactions	2.0 M	21.0 M
Number of Cards	90,748	3.2 M
Number of Fraud Transactions	92,071	17,941
Number of Fraud Cards	46,126	6,355

TABLE II: Overview of the data sets for January 2020.

Overview	Training Set Mar.'19-Dec.'19	Test Set Jan.'20
Number of Transactions	1.9 M	21.3 M
Number of Cards	85,085	3.2 M
Number of Fraud Transactions	90,122	11,245
Number of Fraud Cards	42,198	5,778

TABLE III: Overview of the data sets for February 2020.

Overview	Training Set Apr.'19-Jan.'20	Test Set Feb.'20
Number of Transactions	2.2 M	19.9 M
Number of Cards	90,582	3.1 M
Number of Fraud Transactions	96,908	8,185
Number of Fraud Cards	44,916	4,442

TABLE IV: Overview of the data sets for March 2020.

Overview	Training Set May'19-Feb.'20	Test Set Mar.'20
Number of Transactions	2.2 M	21.3 M
Number of Cards	90,135	3.2 M
Number of Fraud Transactions	93,050	10,758
Number of Fraud Cards	44,716	5,244

IV. EXPERIMENTS

A. Experimental Setup

The experiments were performed using four training sets and test sets incorporating real credit card transactions in a private bank. These four training sets containing transactions 10-month CNP transactions balanced in the card-level before the test month. For example; the transactions from 2019 February to 2019 November (Feb.'19-Nov.'19) and from 2019 March to 2019 December (Mar.'19-Dec.'19), are included in the training sets for December 2019 and January 2020, respectively. For the test sets, we use all CNP transactions in the next month for each training set that are from December 2019 (Dec.'19) to March 2020 (Mar.'20). The details about the data sets are given in Tables I–IV. Also, in each day, there are almost 400 transactions on the average that are reported as fraud or non-fraud, and used for incremental learning. In the experiments, 30% of each training set has been utilized as a validation set, to be used in the training of the detection model.

The proposed system for credit card fraud detection was developed in Python 3.6.8. For the GBT model generation, XGBoost library 0.71 presenting an efficient GBT implementation is used on these data sets. Redis, which is an in-memory data structure store based on a server-client model with TCP sockets, is used for the real-time transaction aggregation.

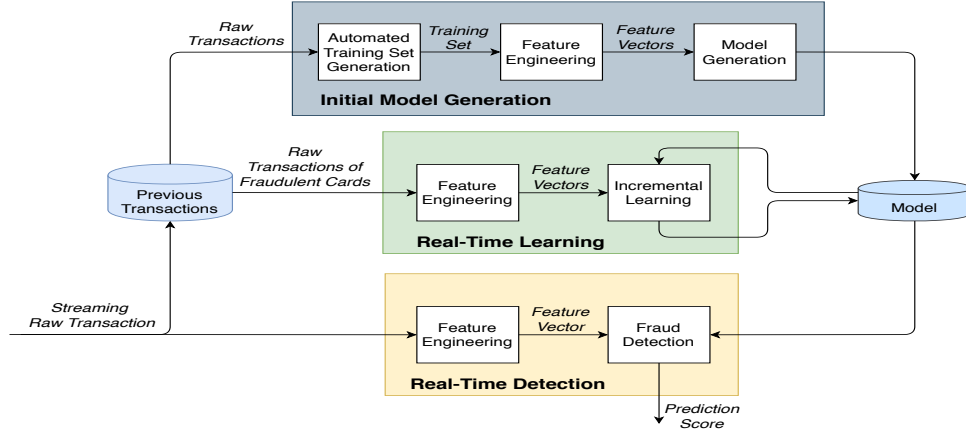


Fig. 1: The flowchart of the fraud detection method with offline training, real-time learning, and credit card fraud detection processes.

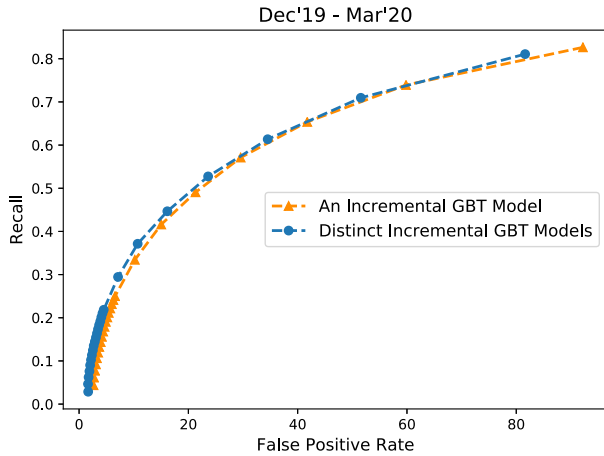


Fig. 2: The overall performances of an incremental learning process during the 4-month period and a distinct learning process for each month in this period.

B. Evaluation Metrics

In the experiments, the performances of the GBT model, card-based incremental GBT model, and retraining based model are evaluated in terms of false positive rate (FPR) and recall. For the learning of all the transactions of the cards, the memory and the computational times consumed by the incremental and retraining models are investigated to compare the models in terms of costs for real-time fraud detection.

C. Results

In the first experiment, the performance of the incremental learning process during the four consecutive months from December 2019 to March 2020 and the overall performance of a distinct incremental learning process for each month in this period are shown in Fig. 2. For each month, a new GBT model is generated with a different training set constructed

by the set construction module, and all the CNP transactions of the related month are incrementally learned by the model. However, the construction requires additional memory and computational cost. Therefore, to avoid the cost, a 4-month incremental learning process is evaluated in this experiment, in which only one training set is constructed for the generation of a GBT model at the beginning of the learning process, and then this model incrementally learned during the four months. This figure demonstrates that the performance of incrementally learning during the 4-month period in which the module is utilized only in the first month, December, has similar performance with the overall performance of the four distinct processes.

In Fig. 3, the results of three models that are GBT, incremental GBT, and retrained GBT models are demonstrated on the 4-month CNP transactions in which the concept drift problem exists only in December 2019. The incremental and retraining processes began with the initial training set constructed for December that is the first month of this period, and all the transactions of the cards reported as fraudulent were incrementally learned, or retrained. For each month, the recall versus FPR performance of the GBT model presented in the previous work is increased by the incremental and retraining models. The results obtained in December demonstrate that both of the models inserting the new transactions into the model have substantially improved the performance of the GBT model, and adapted to the concept drift problem. The AUC score for each test set and the average training time for each day in the testing month are listed in Table V, in which also the average AUC score and average of the average training time are given for each GBT model. The worst AUC score, 0.72379 of the regular GBT model, obtained from the transactions in December including also the drifted ones is improved by the incremental and retrained models, to 0.91053 and 0.91823, respectively. The AUC scores of the incremental model, including the performance of the month with drifting

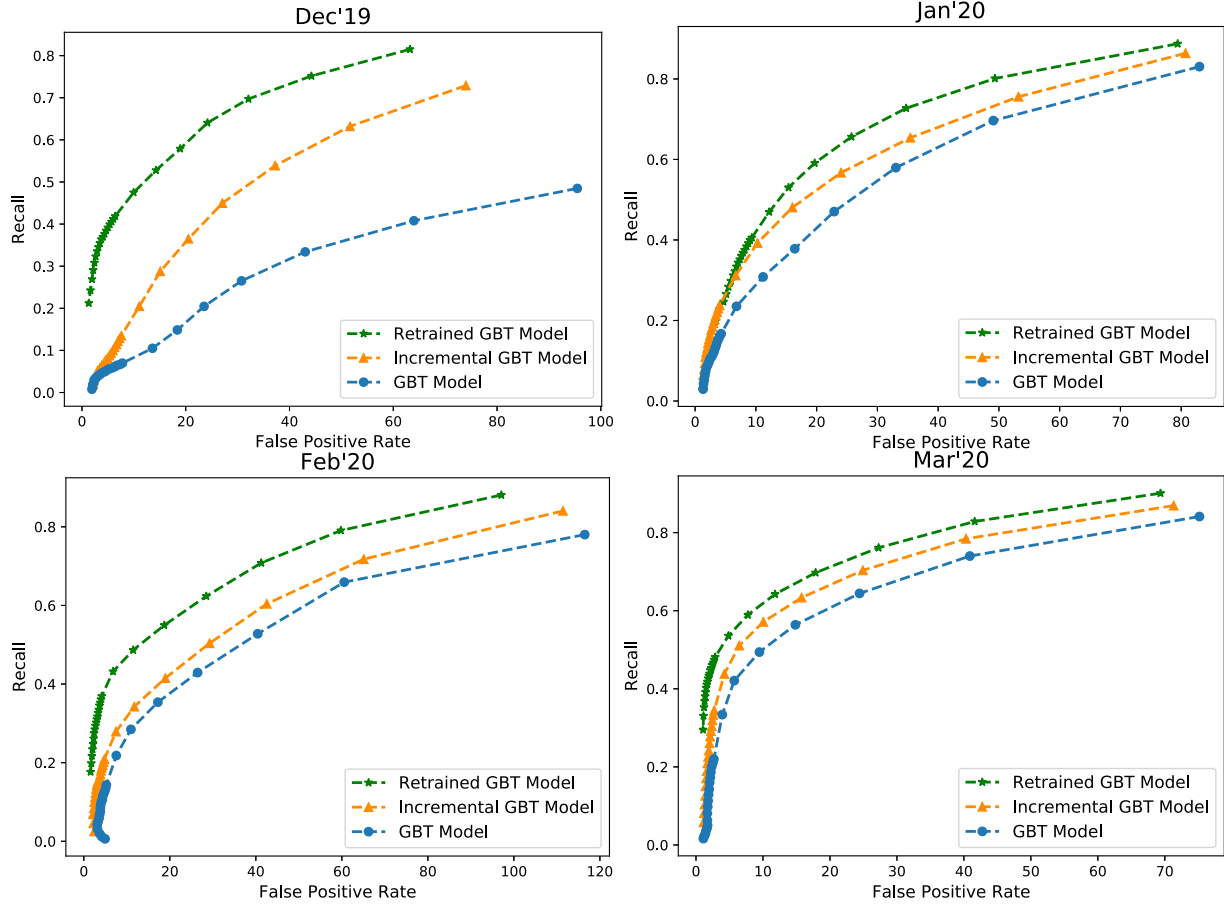


Fig. 3: Credit card fraud detection performances of the GBT model, the retrained GBT model and incremental GBT model

TABLE V: Fraud detection performances of the GBT models on all the CNP transactions of four months.

	GBT		Incremental GBT		Retrained GBT	
Test Set	AUC Score	AUC Score	Training Time [sec.]	AUC Score	Training Time [sec.]	
Dec.'19	0.72379	0.91053	0.384	0.91823	456.825	
Jan.'20	0.97846	0.98136	0.422	0.98474	491.854	
Feb.'20	0.97061	0.98001	0.367	0.98480	574.271	
Mar.'20	0.98272	0.98608	0.471	0.98873	561.980	
Average	0.91390	0.96450	0.411	0.96912	521.232	

transactions are so close to the retrained GBT model. The average incremental training time for each day is 0.418 sec., and only the 10-month transactions of the reported cards, 6147 transactions on average are retained for the training. However, the average computational time of the retraining process in each day is 647 sec., and all the transactions including also the initial set, more than 1.6 million transactions need to be saved. The required resources for retraining are so expensive in real-time learning. Furthermore, in the other months with no drifts, the incremental model has similar performance with the retraining model, therefore, it is decided to use the incremental GBT model that overcomes the drifting problem and improves the regular GBT with low resources.

Furthermore, in the last experiment, it is investigated how

instance-increment based incremental learning causes forgetting the previously learned information. In Fig. 4, the effect of the catastrophic forgetting problem is demonstrated when the daily reported transactions were incrementally learned in December, the month with the drifting problem, and March that is one of the months with perfect performance. Retraining of the model with a new training set including the previous transactions and only these daily transactions may cause a little deterioration or improvement on the performance of the GBT model.

V. CONCLUSION

In real-world streaming data, unexpected changes in the classes and non-class attributes, known as concept drifts can be observed, affecting the performances of machine learning

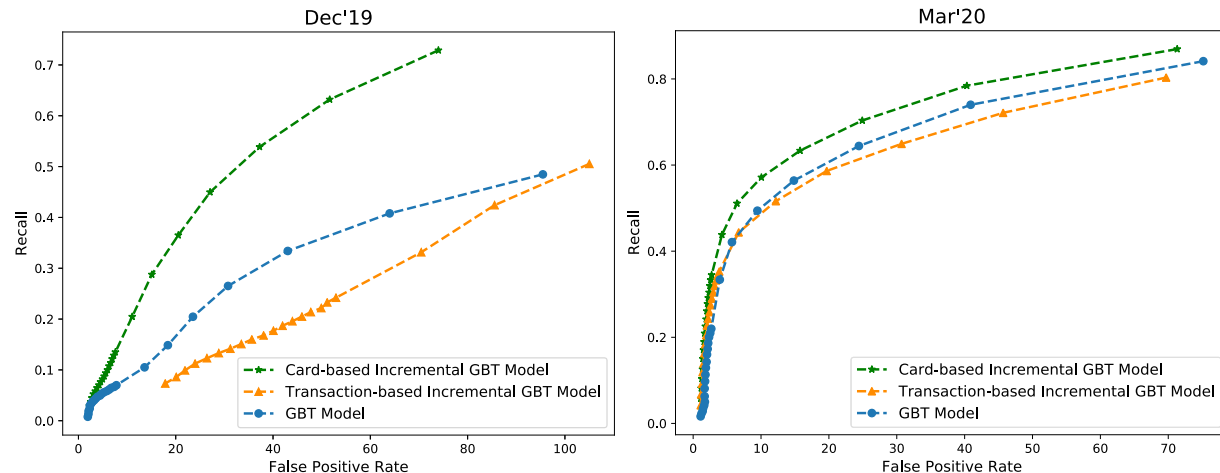


Fig. 4: The performances of GBT model, card-based and transaction-based incremental GBT models.

models. In this work, an incremental learning based approach with Gradient Boosting Trees (GBT) was proposed for the real-time detection of credit card fraud and the adaptation of the concept drift problem that is a major problem affecting the durability of the model. The incremental learning is achieved by card-based increment using the transactions of the cards reported as fraudulent in each day. The significant improvements on the performances of the GBT based method proposed in the previous work were demonstrated, and the catastrophic forgetting that is the most-known problem of incremental learning was investigated by comparing the card- and transaction-based increments in the experiments.

In the future work, new additional aggregated features will be generated and multiple incremental models will be ensemble, that are generated on the transactions of the long- and short-term periods. The incremental GBT model will also be investigated on different real-world fraud detection problems.

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